

# Control Statements in Python

## Selection Statements

### Simple if

```
In [1]: 1 # Example for if statement
        2 # Importance of indentation
        3
        4 # Parenthesis are not required for condition
        5 # True case
        6
        7 if 5<10:
        8     print("5 is less than 10")
        9     print("Second line within the same block")
       10
       11 print("\n This line is out of the if block")
```

5 is less than 10

Second line within the same block

This line is out of the if block

```
In [2]: 1 # Example for if statement
        2 # False case
        3
        4 if 5>10:
        5     print("5 is less than 10")
        6     print("Second line within the same block")
        7
        8 print("\n This line is out of the if block")
```

This line is out of the if block

```
In [4]: 1 age = int (input("Enter your age"))
        2
        3 if age >= 18:
        4     print("You are Eligible to cast vote")
```

Enter your age5

```
In [5]: 1 # Check if a number is positive
2 number = 10
3
4 if number > 0:
5     print("This is a positive number.")
6
7 # If number was -5, nothing would be printed.
```

This is a positive number.

```
In [5]: 1 # Program to print the largest of
2 # the given three numbers.
3 # using Logical operators
4
5 num1 = int(input("Enter First number "))
6 num2 = int(input("Enter Second number "))
7 num3 = int(input("Enter Third number "))
8
9 if num1>num2 and num1>num3:
10     print("num1 is largest")
11
12 if num2>num1 and num2>num3:
13     print("num2 is largest")
14
15 if num3>num1 and num3>num2:
16     print("num3 is largest")
```

Enter First number 10  
Enter Second number 20  
Enter Third number 30  
num3 is largest

## if-else

```
In [7]: 1 # Largest of Two number
2 # Real-time: Comparison in algorithms
3
4 a = int(input("Enter first number: "))
5 b = int(input("Enter second number: "))
6
7 if a > b:
8     print("A is greater")
9 else:
10     print("B is greater")
```

Enter first number: 85  
Enter second number: 10  
A is greater

```
In [9]: 1 # Check pass or fail
        2 # Real-time: Result processing systems
        3
        4 marks = int(input("Enter marks: "))
        5
        6 if marks >= 35:
        7     print("Pass")
        8 else:
        9     print("Fail")
```

Enter marks: 85

Pass

```
In [11]: 1 # if-else Condition
        2 # Program to check whether a number is even or odd.
        3
        4 num = int(input("enter the number"))
        5
        6 if num%2 == 0:
        7     print("Number is even...")
        8 else:
        9     print("Number is odd...")
```

enter the number7

Number is odd...

```
In [15]: 1 # Login System
        2 # Real-time: Authentication systems
        3
        4 username = input("Enter username: ")
        5 password = input("Enter password: ")
        6
        7 if username == "admin" and password == "1234":
        8     print("Login successful")
        9 else:
        10    print("Invalid credentials")
```

Enter username: admin

Enter password: 5618

Login successful

## if-elif-else Ladder

```
In [4]: 1 # if-elif-else ladder
        2 # Determine grade based on score
        3 # Real-time: Academic grading systems
        4
        5 score = int(input("Enter Marks of the student"))
        6
        7 if score >= 90:
        8     print("Grade: A")
        9 elif score >= 80:
       10     print("Grade: B")
       11 elif score >= 70:
       12     print("Grade: C")
       13 else:
       14     print("Grade: D")
```

Enter Marks of the student30  
Grade: D

```
In [6]: 1 # Simple Calculator
        2 # Real-time: Calculator apps
        3
        4 a = int(input("Enter first number: "))
        5 b = int(input("Enter second number: "))
        6 op = input("Enter operator (+, -, *, /): ")
        7
        8 if op == '+':
        9     print(a + b)
       10 elif op == '-':
       11     print(a - b)
       12 elif op == '*':
       13     print(a * b)
       14 elif op == '/':
       15     print(a / b)
       16 else:
       17     print("Invalid operator")
```

Enter first number: 10  
Enter second number: 20  
Enter operator (+, -, \*, /): @  
Invalid operator

In [ ]:

```
1  # Level: Advanced
2  # Temperature Conversion Program
3  # Menu-based: Convert between Celsius, Fahrenheit, and Kelvin
4
5  # Show menu options
6  print("Temperature Conversion Menu:")
7  print("1. Celsius to Fahrenheit")
8  print("2. Celsius to Kelvin")
9  print("3. Fahrenheit to Celsius")
10 print("4. Fahrenheit to Kelvin")
11 print("5. Kelvin to Celsius")
12 print("6. Kelvin to Fahrenheit")
13
14 # Ask user to choose an option
15 choice = int(input("Enter your choice (1-6): "))
16
17 # Ask user to enter temperature value
18 temp = float(input("Enter the temperature value: "))
19
20 # Perform conversion based on choice
21 if choice == 1:
22     # Celsius to Fahrenheit:  $(C \times 9/5) + 32$ 
23     result = (temp * 9/5) + 32
24     print("Result:", result, "°F")
25
26 elif choice == 2:
27     # Celsius to Kelvin:  $C + 273.15$ 
28     result = temp + 273.15
29     print("Result:", result, "K")
30
31 elif choice == 3:
32     # Fahrenheit to Celsius:  $(F - 32) \times 5/9$ 
33     result = (temp - 32) * 5/9
34     print("Result:", result, "°C")
35
36 elif choice == 4:
37     # Fahrenheit to Kelvin:  $(F - 32) \times 5/9 + 273.15$ 
38     result = (temp - 32) * 5/9 + 273.15
39     print("Result:", result, "K")
40
41 elif choice == 5:
42     # Kelvin to Celsius:  $K - 273.15$ 
43     result = temp - 273.15
44     print("Result:", result, "°C")
45
46 elif choice == 6:
47     # Kelvin to Fahrenheit:  $(K - 273.15) \times 9/5 + 32$ 
48     result = (temp - 273.15) * 9/5 + 32
49     print("Result:", result, "°F")
50
51 else:
52     # If user enters invalid choice
53     print("Invalid choice! Please select between 1-6.")
```



In [ ]:

```
1  # Level: Advanced
2  # Mini Unit Converter
3  # Converts between Length, Weight, and Time
4  # Menu-based
5
6  # Show main menu
7  print("Unit Converter Menu:")
8  print("1. Length Conversion")
9  print("2. Weight Conversion")
10 print("3. Time Conversion")
11
12 # Ask user to choose conversion type
13 choice = int(input("Enter your choice (1-3): "))
14
15 # Length Conversion
16 if choice == 1:
17     print("\nLength Conversion Options:")
18     print("1. Meters to Kilometers")
19     print("2. Kilometers to Meters")
20     print("3. Centimeters to Meters")
21     print("4. Meters to Centimeters")
22
23     sub_choice = int(input("Enter your choice (1-4): "))
24     value = float(input("Enter the value: "))
25
26     if sub_choice == 1:
27         result = value / 1000
28         print("Result:", result, "km")
29     elif sub_choice == 2:
30         result = value * 1000
31         print("Result:", result, "m")
32     elif sub_choice == 3:
33         result = value / 100
34         print("Result:", result, "m")
35     elif sub_choice == 4:
36         result = value * 100
37         print("Result:", result, "cm")
38     else:
39         print("Invalid choice!")
40
41 # Weight Conversion
42 elif choice == 2:
43     print("\nWeight Conversion Options:")
44     print("1. Kilograms to Grams")
45     print("2. Grams to Kilograms")
46     print("3. Kilograms to Pounds")
47     print("4. Pounds to Kilograms")
48
49     sub_choice = int(input("Enter your choice (1-4): "))
50     value = float(input("Enter the value: "))
51
52     if sub_choice == 1:
53         result = value * 1000
54         print("Result:", result, "g")
55     elif sub_choice == 2:
56         result = value / 1000
57         print("Result:", result, "kg")
```

```
58     elif sub_choice == 3:
59         result = value * 2.20462
60         print("Result:", result, "lb")
61     elif sub_choice == 4:
62         result = value / 2.20462
63         print("Result:", result, "kg")
64     else:
65         print("Invalid choice!")
66
67 # Time Conversion
68 elif choice == 3:
69     print("\nTime Conversion Options:")
70     print("1. Seconds to Minutes")
71     print("2. Minutes to Seconds")
72     print("3. Hours to Minutes")
73     print("4. Minutes to Hours")
74
75     sub_choice = int(input("Enter your choice (1-4): "))
76     value = float(input("Enter the value: "))
77
78     if sub_choice == 1:
79         result = value / 60
80         print("Result:", result, "minutes")
81     elif sub_choice == 2:
82         result = value * 60
83         print("Result:", result, "seconds")
84     elif sub_choice == 3:
85         result = value * 60
86         print("Result:", result, "minutes")
87     elif sub_choice == 4:
88         result = value / 60
89         print("Result:", result, "hours")
90     else:
91         print("Invalid choice!")
92
93 else:
94     print("Invalid choice! Please select between 1-3.")
```



## match-case

```
In [8]: 1 # A simple variable representing a user's role
2 user_role = "admin"
3
4 # Implementing match-case directly in the main flow
5 match user_role:
6     case "admin":
7         print("Full access granted: You can edit, delete, and add users.")
8     case "editor":
9         print("Partial access: You can edit and publish content.")
10    case "guest":
11        print("Limited access: You can only view content.")
12    case _:
13        print("Access denied: Role not recognized.")
14
```

Full access granted: You can edit, delete, and add users.

```
In [2]: 1 # Real-Life Example: HTTP Status Codes
2
3 def handle_response(status_code):
4     match status_code:
5         case 200:
6             return "Success"
7         case 404:
8             return "Not Found"
9         case 500:
10            return "Server Error"
11        case _:
12            return "Unknown status code"
13
14 print(handle_response(200))
15 print(handle_response(404))
16 print(handle_response(302))
```

Success  
Not Found  
Unknown status code

## Nested if-else

```
In [12]: 1 # Python Nested if Example
          2
          3 num = int(input("Enter a number: "))
          4 if num >= 0:
          5     if num== 0:
          6         print("Zero")
          7     else:
          8         print("Positive number")
          9 else:
         10     print("Negative number")
```

Enter a number: 5

Positive number